Extending class imbalance mitigation benchmarks to more datasets

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Abstract

A prior study benchmarked class-imbalance mitigation methods on TCGA-UT, a pan-cancer whole-slide-image cohort with a native, moderately imbalanced slide distribution (Queisler, 2026b). This report extends the same benchmark machinery to three additional datasets, keeping the frozen Virchow2 representation, a bounded per-slide evidence budget, and the patient-disjoint evaluation protocol fixed, and asks whether the method comparison remains meaningful when the data regime changes. The datasets are chosen to bracket the imbalance spectrum and to vary organ, label type, and annotation form: BRACS, an organ-specific H&E breast-lesion cohort with a mildly imbalanced native seven-subtype distribution (head-to-tail ≈2.3:1); CAMELYON16, a sentinel lymph-node metastasis cohort whose mask-derived tumor/normal patch distribution is severely imbalanced (≈39:1); and PANDA, a large prostate-biopsy cohort whose native six-class ISUP grade distribution is again only mildly imbalanced (≈2.4:1) but whose task is ordinal grading, evaluated jointly with a provider-consistent tile-level cancer/benign target derived from its masks. Both patch-feature and whole-slide-image feature-bag method families are compared, each within its own prediction regime, and a plain cross-entropy no-mitigation baseline (CE, MIL CE) is added so mitigation gains are measured against an unweighted reference. On BRACS, OKO and CFAL lead the patch regime by a few points of balanced accuracy while every other method—including ProGAN augmentation—is statistically indistinguishable from the unweighted baseline, and in the WSI-bag regime the unweighted MIL CE baseline is the strongest method, with no mitigation method improving on it; both regimes remain substantially miscalibrated after temperature scaling. On CAMELYON16, despite an imbalance an order of magnitude more severe, every method in both regimes reaches near-ceiling discrimination (balanced accuracy 0.96–0.99) and near-ideal calibration, and no mitigation method separates from the unweighted baseline. BRACS and CAMELYON16 together show that imbalance severity alone predicts neither task difficulty nor the value of mitigation: what governs the headroom for imbalance mitigation under a frozen-feature benchmark is the separability of the classes in the representation, not the head-to-tail ratio. PANDA confirms this within one cohort whose two targets are both only mildly imbalanced: the separable tile-level cancer-detection regime is comparatively easy (balanced accuracy ≈0.85) while the entangled slide-level six-class ISUP-grading regime stays hard (macro F1 ≈0.58), and the unweighted baseline is competitive in both.

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1 Introduction

A companion study compared class-imbalance mitigation methods under the native, moderately imbalanced slide distribution of TCGA-UT (head-to-tail ≈28:1) across 32 cancer types (Queisler, 2026b; Tomczak et al., 2015; Komura et al., 2022). That benchmark fixes a frozen Virchow2 encoder (Vorontsov et al., 2024; Zimmermann et al., 2024), a bounded per-slide evidence budget, and patient-disjoint splits, so that the only thing being compared is how each mitigation method reshapes minibatch exposure or the per-example loss. The natural next question is external validity: does the same benchmark machinery stay meaningful when it is pointed at a different dataset, with a different organ, a different label set, and a different native class distribution?

This report takes that step by extending the benchmark to three further datasets on their *native* class distributions, chosen to span the imbalance spectrum and to vary organ, label type, and annotation form. The design is deliberately conservative: the representation family (frozen Virchow2), the bounded per-slide evidence budget, the patient-disjoint split protocol, the method families, and the validation-based model-selection rule are all carried over unchanged, and only the dataset is swapped. A dataset therefore acts as a check on how far the benchmark generalizes rather than as a new experimental variable.

The three datasets—BRACS (breast-lesion subtyping, mild native imbalance ≈2.3:1), CAMELYON16 (sentinel lymph-node metastasis detection, severe patch-level imbalance ≈39:1), and PANDA (ordinal prostate ISUP grading, mild imbalance ≈2.4:1, plus a tile-level cancer/benign target)—are introduced in full, with their clinical background, provenance, and class distributions, in the companion datasets report (Queisler, 2026a). They vary organ, label type, and annotation form, so that together they test whether the method comparison holds across mild and severe imbalance and across a breast-lesion subtype task, a lymph-node detection task, and a prostate grading task—the variation that stresses a benchmark's generality.

Both regimes from the companion study are carried over. Patch-feature methods include cross-entropy, class weighting, balanced sampling, focal loss, Scholz-style metric losses, CFAL, OKO set learning, and ProGAN synthetic augmentation (Scholz et al., 2024). WSI-bag methods include cross-entropy, class weighting, balanced sampling, focal loss, RankMix, SC-MIL, and MDE-MIL (Chen and Lu, 2023; Juyal et al., 2024; Ling et al., 2025). Patch and WSI-bag methods are reported separately because they consume different prediction units and answer regime-specific questions.

2 Datasets

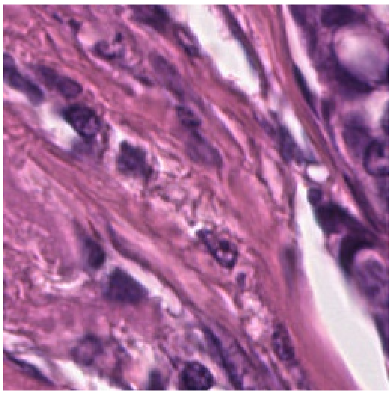
The benchmark evaluates each dataset on its *native* class distribution: the observed per-class support is measured after a patient-disjoint preparation and used directly, without any re-sampling. The clinical background, provenance, and full native class distributions of BRACS, CAMELYON16, and PANDA are given in the companion datasets report (Queisler, 2026a); this section states only what the benchmark consumes—the target labels, the per-slide evidence budget, and the dataset-specific preparation choices that the results depend on. Further datasets can be added under the same protocol.

2.1 BRACS

The benchmark uses BRACS's seven-subtype target—normal (N), pathological benign (PB), usual ductal hyperplasia (UDH), flat epithelial atypia (FEA), atypical ductal hyperplasia (ADH), ductal carcinoma in situ (DCIS), and invasive carcinoma (IC)—because it is the level at which class imbalance is most relevant: the atypical FEA and ADH lesions are clinically important precancers that are scarcer than most benign or malignant samples, so collapsing to the three parent lesion types would hide part of the rare-class problem (Brancati et al., 2022; Queisler, 2026a). Patient identifiers from the BRACS summary sheet define the split unit, and the prepared benchmark enforces patient-disjoint train, validation, and test partitions across three seeds, matching the leakage-control standard used for TCGA-UT.

BRACS evidence is derived from the ROI set. Each annotated ROI is tiled deterministically into 256×256 RGB patches, and each slide's pooled ROI tiles are then capped at the empirical median number of available tiles per slide (360)—the same data-driven per-slide evidence budget used for CAMELYON16 and PANDA—with slides below the median kept in full. Patch methods consume one frozen Virchow2 feature per ROI tile, represented as the concatenation of the CLS token and the mean of the patch tokens (2,560-dimensional vectors). The WSI-bag pipeline consumes grouped ROI-tile features as bounded bags, preserving the same patch-versus-bag comparison while avoiding an additional tissue-detection step on full WSIs. This choice prioritizes label fidelity: all training instances originate from pathologist-selected lesion regions rather than from unlabeled background tissue.

After patch extraction, 387 of the 547 published WSIs contribute at least one annotated ROI tile; the remaining slides have no tiled ROI annotations available. The prepared benchmark covers 151 patients, the full 4,539 ROI metadata records, and 99,352 extracted patches (Table 1). The seven-subtype WSI support spans PB (196 slides) at the head to DCIS (87 slides) at the tail, a native head-to-tail ratio of approximately 2.25:1 (entropy-based imbalance $1 - H_{\text{norm}} \approx 0.02$; Figure 1). This is mild imbalance relative to the pan-cancer TCGA-UT cohort, but it reflects the true clinical distribution of breast-lesion subtypes and is used as-is.



Property	Value
Patients	151
WSIs with ROI tiles	387
ROI metadata rows	4,539
ROI tiles	99,352
WSI-bag size (median tiles)	360
Subtype labels	N, PB, UDH, FEA, ADH, DCIS, IC
WSI support range	87–196
WSI imbalance ratio	≈2.3:1

Table 1: BRACS dataset properties used by the native benchmark, after patient-disjoint preparation and ROI tiling, alongside an example 256×256 ROI tile. Each tile is one frozen Virchow2 feature consumed by the patch regime and a candidate instance in the WSI bag.

2.2 CAMELYON16

CAMELYON16 provides binary slide labels—each sentinel lymph-node section is either *normal* or contains *tumor* (metastasis)—with expert annotations supplied as pixel-level tumor masks rather than region-of-interest crops (Ehteshami Bejnordi et al., 2017; Litjens et al., 2018; Queisler, 2026a). Holding the frozen Virchow2 representation and the patient-disjoint protocol fixed while changing the organ, the label granularity, and, above all, the severity of the native imbalance, it is the deliberate severe counterpart to BRACS's mild seven-subtype distribution.

The benchmark uses 398 of the 399 published slides, comprising 160 tumor and 238 normal slides after one slide without usable tile evidence is excluded. Each slide is its own patient (case), so slide-disjoint train, validation, and test partitions across three seeds coincide with patient-disjoint splits. CAMELYON16 is evaluated in the same two regimes as BRACS, but with regime-specific labels. In the patch regime, each tile is labelled tumor or normal by intersecting it with the published tumor mask; because metastatic regions occupy only a small fraction of the sectioned tissue, tumor tiles are scarce and the native patch distribution is severely imbalanced

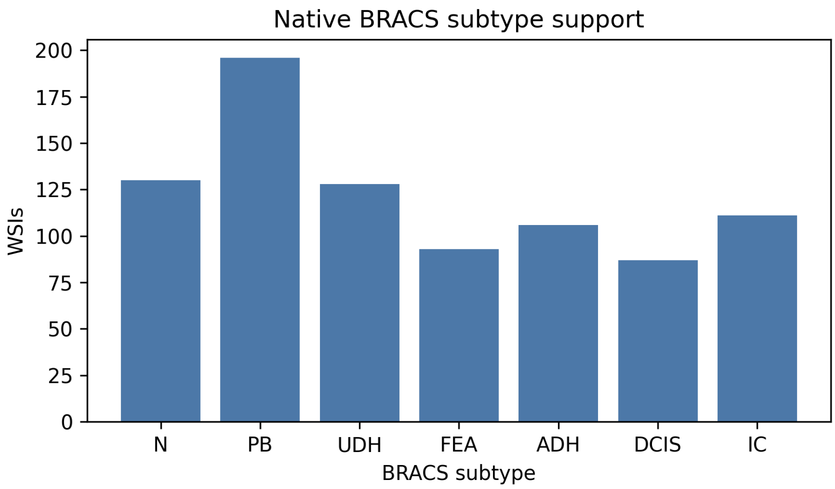
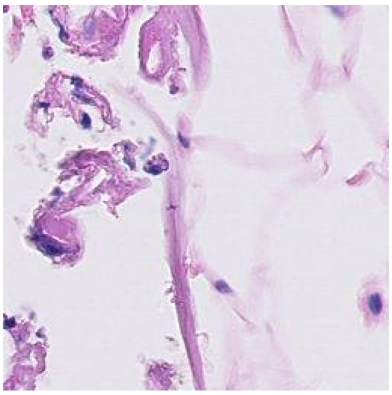


Figure 1: Native BRACS WSI support by seven-subtype label after patient-disjoint preparation. The distribution is used directly as the benchmark's native class-imbalance setting.

(head-to-tail ≈39:1, entropy-based imbalance $1 - H_{\text{norm}} \approx 0.83$; Table 2 and Figure 2). In the WSI-bag regime, each slide is a single bag carrying its slide-level tumor/normal label, which recovers the canonical CAMELYON16 metastasis-detection task.

Two dataset-specific choices follow from this design. First, the 20 tumor slides whose annotations are not exhaustive—so that unannotated tumor could be mislabelled as normal—are excluded from the patch regime but retained in the WSI-bag regime, where only the reliable slide-level label is used. Second, the per-slide evidence budget is set data-driven rather than fixed: the WSI-bag size is the empirical median number of available tissue tiles per slide, which for CAMELYON16 is 7,874 tiles. This is two orders of magnitude larger than the 30-tile budget that the tile-sparse TCGA-UT and BRACS cohorts admit, and reflects that a whole lymph-node section yields far more tissue than a single lesion ROI. Patch evidence is otherwise the same frozen Virchow2 2,560-dimensional embedding used throughout the benchmark, computed once per tile and shared by both regimes.



Property	Value
WSIs (tumor / normal)	160 / 238
Patch-labelled WSIs	378
WSI-bag size (median tiles)	7,874
Patch labels	tumor, normal
Tumor / normal patches	59,672 / 2,340,040
Patch imbalance ratio	≈39.2:1

Table 2: CAMELYON16 dataset properties used by the native benchmark, after patient-disjoint preparation and mask-based patch labelling, alongside an example 256×256 tumor tile. The WSI-bag size is the data-driven median of available $20\times$ tissue tiles per slide.

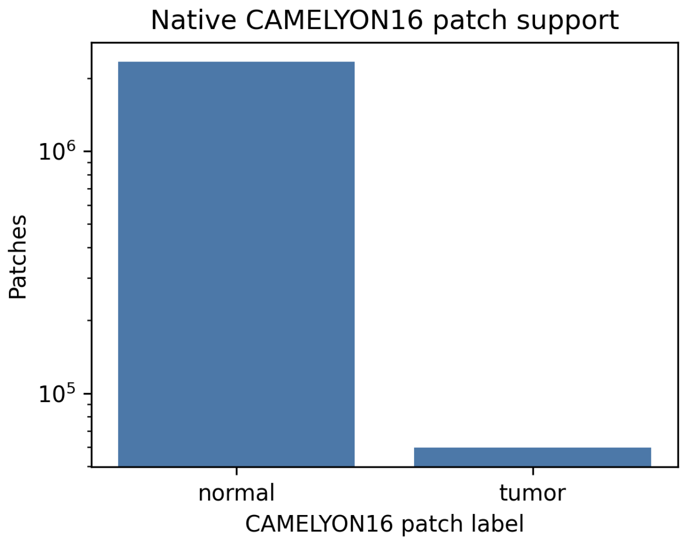


Figure 2: Native CAMELYON16 patch support by tumor/normal label after patient-disjoint preparation and mask-based labelling (log scale). Tumor tiles are roughly 39 times scarcer than normal tiles, giving the benchmark its severe native imbalance setting.

2.3 PANDA

PANDA is a cohort of prostate core-needle biopsies graded on the ordinal six-class ISUP scale, where grade 0 is benign and grades 1–5 are increasingly aggressive cancer grade groups derived from Gleason morphology (Bulten et al., 2022; Queisler, 2026a). It differs from BRACS and CAMELYON16 along every axis a generality check exercises: a third organ (prostate), an ordinal label rather than nominal subtypes or a binary flag, and a built-in source of heterogeneity, because the slides come from two institutions—Radboud University Medical Center and Karolinska Institute—that differ in staining, scanning, and annotation convention.

The WSI-bag regime uses the canonical PANDA target, the ISUP grade group. Because most biopsies are benign or low grade, the native ISUP distribution is only mildly imbalanced (head-to-tail ≈2.4:1, entropy-based imbalance $1 - H_{\text{norm}} \approx 0.04$), placing PANDA near BRACS on the imbalance spectrum but with a coarser, ordinal label. The two providers grade differently—Karolinska biopsies concentrate in the low grades while Radboud biopsies spread more evenly across the scale—so provider identity is a genuine distribution-shift axis carried inside a single native cohort.

The patch regime uses a provider-consistent binary cancer/benign target derived from the pixel masks. PANDA's expert masks are region annotations whose value semantics are institution-dependent: Radboud masks distinguish stroma, benign epithelium, and Gleason patterns 3–5, whereas Karolinska masks mark only benign versus cancerous tissue. Collapsing both to a common cancer/benign distinction is the only tile label the two providers share, so each 256×256 tile is labelled *cancer* when at least half of its mask cell carries a cancer value and *benign* otherwise. The two regimes therefore answer regime-specific questions with regime-specific labels—slide-level ordinal grading against tile-level cancer detection—mirroring the split already used for CAMELYON16.

Two dataset-specific choices follow from this design. First, tile evidence is generated from the whole-slide images: each biopsy is read at the 20× pyramid level, restricted to tissue tiles by a stain-intensity filter, and cut deterministically into 256×256 RGB patches, each embedded once as a frozen Virchow2 feature. Second, to keep frozen-feature extraction tractable on a cohort an order of magnitude larger than the other two, the benchmark draws a stratified subset of approximately 2,000 biopsies, sampled to preserve the joint distribution of ISUP grade and provider; the per-slide evidence budget is again data-driven, set to the empirical median number of tissue tiles per biopsy (Table 3 and Figure 3). Biopsies whose masks are unavailable keep their slide-level ISUP label for the WSI-bag regime but are excluded from the mask-labelled patch regime, matching the exhaustiveness handling used for CAMELYON16.

Property	Value
WSIs (biopsies)	2,000
Patch-labelled WSIs	1,984
WSI-bag size (median tiles)	421
WSI labels	ISUP 0–5
WSI imbalance ratio	≈2.4:1
Patch labels	benign, cancer
Benign / cancer patches	580,531 / 122,751
Patch imbalance ratio	≈4.7:1

Table 3: PANDA dataset properties used by the native benchmark, after stratified subset selection, patient-disjoint preparation, and mask-based patch labelling. The WSI-bag size is the data-driven median of available 20× tissue tiles per biopsy.

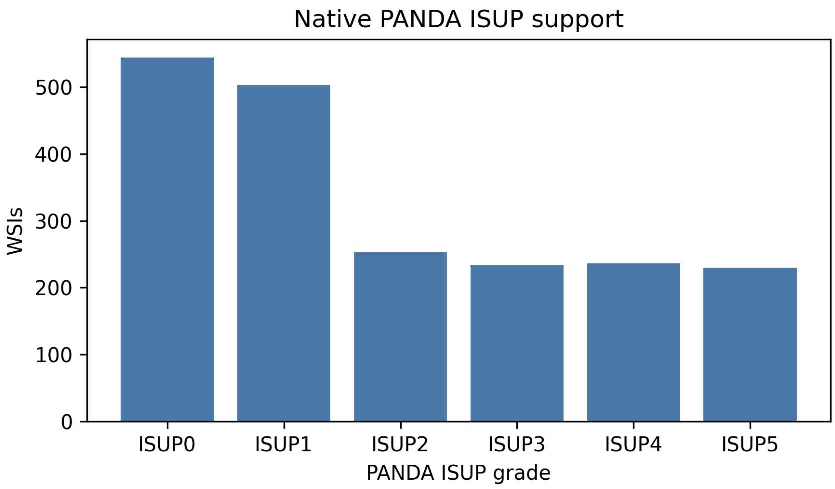


Figure 3: Native PANDA WSI support by six-class ISUP grade after stratified subset selection and patient-disjoint preparation. The distribution is used directly as the benchmark's native class-imbalance setting.

3 Methods

The method taxonomy follows the companion benchmark (Queisler, 2026b). Cross-entropy is the no-mitigation reference. Data-level methods change minibatch exposure through balanced sampling or synthetic augmentation. Algorithm-level methods change the per-example loss through class weighting, focal loss, metric-derived objectives, prototype-affinity objectives, or OKO set supervision. WSI-specific methods change how slide bags are represented or mixed.

Full method definitions follow Queisler (2026b) and Scholz et al. (2024). RankMix, SC-MIL, and MDE-MIL are WSI-bag methods whose mechanisms operate on bags rather than isolated patch embeddings (Chen and Lu, 2023; Juyal et al., 2024; Ling et al., 2025). The plain cross-entropy baselines (CE for the patch regime, MIL CE for the WSI-bag regime) apply uniform

Family	Methods	Regime
No-mitigation baseline	CE, MIL CE	Patch, WSI bag
Data-level sampling and augmentation	Balanced sampling, RankMix	Patch, WSI bag
Algorithm-level losses	Weighted CE, Focal loss, CFAL	Patch (CFAL); Patch and WSI bag (others)
Hybrid sampling–loss	CE + soft F1, CE + soft MCC, OKO, MDE-MIL	Patch (first three, OKO); WSI bag (MDE-MIL)
Synthetic data generation	ProGAN augmentation	Patch
Representation and architecture	SC-MIL	WSI bag

Table 4: Method families evaluated on the native benchmarks (BRACS, CAMELYON16, and PANDA), following the taxonomy of Queisler (2026b). CE and MIL CE are unweighted cross-entropy baselines, added so mitigation gains are measured against a genuine no-mitigation reference. Patch and WSI-bag methods are compared only within their own input regime.

class weights and no resampling, so any improvement from a mitigation method is measured relative to doing nothing.

4 Evaluation Protocol

The benchmark evaluates each dataset on its native class distribution. Hyperparameters are selected independently for every method: for each candidate configuration the validation macro F1 is averaged over the three patient-disjoint seeds, and the configuration with the highest mean is chosen, with ties broken by mean validation balanced accuracy. A single configuration is therefore fixed per method (shared across seeds). Selected configurations are evaluated on the fixed test split, and results are reported as mean ± standard deviation across seeds. Macro F1 is the primary discrimination metric because it weights classes equally; balanced accuracy and accuracy are reported as complementary discrimination metrics.

The comparison is intentionally regime-specific. Patch-feature methods are compared only with other patch-feature methods, and WSI-bag methods are compared only with other WSI-bag methods. The two regimes share the patient-disjoint splits, the frozen Virchow2 representation family, and a bounded per-slide evidence budget, but they differ in prediction unit and aggregation mechanism. The evidence budget itself is dataset-specific but follows a common data-driven rule: each dataset caps its per-slide bag at the empirical median number of available tiles per slide—360 for BRACS, 7,874 for CAMELYON16, and 421 tissue tiles per biopsy for PANDA. These budgets differ by two orders of magnitude because a whole lymph-node section yields far more tissue than a single breast-lesion ROI set or prostate core. The label used within a regime is also dataset-specific where the two prediction units diverge: for PANDA the WSI-bag regime carries the six-class ISUP grade while the patch regime carries the mask-derived binary cancer/benign label, so the same biopsy contributes an ordinal grading target and a tile-level detection target.

Calibration is evaluated alongside discrimination using expected calibration error (ECE; Naeini et al., 2015; Guo et al., 2017). For each method and seed, a single temperature is fit by minimizing negative log likelihood on the validation split and applied unchanged to the test score matrix before softmax (Guo et al., 2017). Tables 7 and 8 report ECE before and after temperature scaling with the fitted temperature.

5 Results

Each dataset is evaluated on its own native distribution, with the patch-feature and WSI-bag regimes reported separately. Scores are specific to each dataset's task and are not directly comparable across datasets or to the 32-class TCGA-UT benchmark, because the label set, task difficulty, and cohort size all differ. The three datasets turn out to span the difficulty–imbalance plane—BRACS is mildly imbalanced but hard, CAMELYON16 severely imbalanced but easy, and PANDA spans both difficulty extremes within one cohort whose two targets are both mildly imbalanced—and it is this spread, rather than any dataset alone, that tests the benchmark's generality.

5.1 BRACS

The patch-feature and WSI-bag regimes are reported separately (Tables 5 and 6). The absolute discrimination and calibration numbers are markedly lower than on TCGA-UT for two concrete reasons: fine-grained breast-lesion subtyping (distinguishing e.g. FEA, ADH, and DCIS) is intrinsically harder than pan-cancer type classification, and the native breast-only training pool is far smaller than the large constructed pan-cancer pools used for TCGA-UT, so each method has both a harder decision boundary and less data to fit it. This is consistent with the external literature: even recent task-specific BRACS models—purpose-built graph and transformer architectures trained end-to-end for this task—report only mid-60% weighted F1 on seven-subtype ROI classification (HACT-Net 61.5%, ScoreNet 64.4%), with the atypical UDH and ADH subtypes still around 45% F1 (Pati et al., 2022; Stegmüller et al., 2023). A frozen-feature benchmark under a bounded evidence budget should therefore not be expected to approach TCGA-UT-level discrimination; the binding limitation is fine-grained lesion morphology and limited task-specific representation learning, not class imbalance alone.

Method	Accuracy	Balanced accuracy	Macro F1
CFAL	0.589 ± 0.027	0.497 ± 0.042	0.490 ± 0.038
OKO	0.565 ± 0.054	0.502 ± 0.052	0.488 ± 0.049
CE + soft F1	0.547 ± 0.034	0.480 ± 0.036	0.468 ± 0.033
CE + soft MCC	0.545 ± 0.043	0.472 ± 0.047	0.463 ± 0.044
Balanced sampling	0.552 ± 0.045	0.470 ± 0.043	0.461 ± 0.045
ProGAN augmentation	0.549 ± 0.049	0.460 ± 0.045	0.457 ± 0.049
Focal loss	0.533 ± 0.041	0.460 ± 0.037	0.452 ± 0.036
CE	0.548 ± 0.045	0.451 ± 0.056	0.451 ± 0.055
Weighted CE	0.531 ± 0.045	0.461 ± 0.040	0.450 ± 0.042

Table 5: BRACS native patch-feature test results (accuracy, balanced accuracy, and macro F1) for validation-selected configurations, mean $\pm$ standard deviation across patient-disjoint seeds.

In the patch regime, OKO and CFAL are the only methods that visibly separate from the unweighted CE baseline, leading it by roughly five balanced-accuracy points while remaining close to each other (Table 5). The rest of the field falls inside one standard deviation of CE, including Weighted CE, focal loss, the soft metric losses, balanced sampling, and ProGAN augmentation. ProGAN is especially informative because it lands essentially on the CE baseline, suggesting that BRACS's mild native imbalance does not create the tail-scarcity regime in which synthetic minority augmentation helped on TCGA-UT. The main result is therefore simple: on BRACS patch features, only OKO and CFAL show a measurable lift over doing nothing.

Method	Accuracy	Balanced accuracy	Macro F1
MIL CE	0.672 ± 0.020	0.524 ± 0.103	0.508 ± 0.045
Weighted CE	0.667 ± 0.029	0.510 ± 0.093	0.501 ± 0.055
Focal loss	0.691 ± 0.029	0.490 ± 0.032	0.498 ± 0.033
MDE-MIL	0.635 ± 0.033	0.489 ± 0.093	0.480 ± 0.051
Balanced sampling	0.635 ± 0.022	0.449 ± 0.022	0.462 ± 0.023
RankMix	0.603 ± 0.104	0.437 ± 0.087	0.446 ± 0.075
SC-MIL	0.539 ± 0.092	0.367 ± 0.083	0.383 ± 0.086

Table 6: BRACS native WSI-bag test results for validation-selected configurations, mean ± standard deviation across patient-disjoint seeds.

In the WSI-bag regime, the unweighted MIL CE baseline has the highest balanced accuracy and macro F1, with Weighted CE close behind and focal loss trading balanced performance for the highest overall accuracy (Table 6). No mitigation method improves on MIL CE on the balanced metrics, and the bag-composition methods are weaker on this dataset, with SC-MIL the clearest failure case. The large balanced-accuracy standard deviations in the table are not a typographical error: the BRACS WSI test sets are small after patient-disjoint splitting, and the class supports shift noticeably across the three seeds, so a few corrected or missed slides in rare subtypes move balanced accuracy by several points. The top ordering should therefore be read as unstable, but the qualitative conclusion is robust: plain MIL CE is at least as good as every dedicated bag-composition and resampling method on BRACS, exactly the failure mode a no-mitigation baseline is included to expose.

Method	ECE	ECE+TS	T
CE	0.354 ± 0.039	0.031 ± 0.010	5.812 ± 0.409
Weighted CE	0.367 ± 0.039	0.038 ± 0.017	5.752 ± 0.486
Focal loss	0.286 ± 0.023	0.030 ± 0.017	3.459 ± 0.405
CFAL	0.417 ± 0.026	0.049 ± 0.013	0.079 ± 0.010
CE + soft F1	0.405 ± 0.030	0.070 ± 0.027	8.532 ± 1.329
CE + soft MCC	0.366 ± 0.039	0.029 ± 0.011	6.582 ± 0.723
Balanced sampling	0.347 ± 0.036	0.021 ± 0.011	5.823 ± 0.601
OKO	0.045 ± 0.028	0.039 ± 0.014	0.877 ± 0.149
ProGAN augmentation	0.358 ± 0.041	0.039 ± 0.018	6.158 ± 0.677

Table 7: BRACS native patch-feature calibration: ECE before (raw) and after (TS) temperature scaling, with the fitted temperature T, mean ± standard deviation across seeds.

Probability Calibration. In the patch regime, raw (pre-scaling) miscalibration is severe for most methods: ECE ranges from about 0.29 to 0.42, requiring large temperature corrections ($T \approx 3$–9) to reach post-scaling ECE of 0.02–0.07. OKO is the exception, with by far the lowest raw ECE (0.045) and a fitted temperature close to one ($T \approx 0.88$), consistent with its behaviour on the companion TCGA-UT benchmark. CFAL again shows the CFAL-specific pattern of a near-degenerate low temperature ($T \approx 0.08$) alongside a high raw ECE (0.417), echoing its affinity-based outputs being nearly flat prior to scaling, which scaling nonetheless corrects to 0.049. After scaling, balanced sampling is the best-calibrated method (ECE 0.021), followed closely by CE + soft MCC (0.029), focal loss (0.030), and the unweighted CE baseline

Method	ECE	ECE+TS	T
MIL CE	0.237 ± 0.012	0.165 ± 0.040	3.092 ± 0.322
Weighted CE	0.247 ± 0.051	0.163 ± 0.045	2.928 ± 0.343
Focal loss	0.164 ± 0.066	0.136 ± 0.010	1.977 ± 0.075
Balanced sampling	0.257 ± 0.064	0.168 ± 0.052	2.861 ± 0.314
RankMix	0.151 ± 0.074	0.144 ± 0.017	1.298 ± 0.108
SC-MIL	0.302 ± 0.097	0.168 ± 0.014	2.421 ± 0.242
MDE-MIL	0.277 ± 0.017	0.151 ± 0.010	2.083 ± 0.326

Table 8: BRACS native WSI-bag calibration: ECE before and after temperature scaling, with the fitted temperature T. Columns as in Table 7.

(0.031); CE + soft F1 is the worst (0.070, $T \approx 8.5$). ProGAN augmentation calibrates like the unweighted baselines (raw ECE 0.358, $T \approx 6.2$, post-scaling ECE 0.039), showing no distinctive miscalibration pattern. In the WSI-bag regime, calibration is uniformly worse than in the patch regime and than on the companion TCGA-UT benchmark: post-scaling ECE ranges from 0.136 (focal loss, the best-calibrated method) to 0.168 (balanced sampling and SC-MIL, tied for worst), with the unweighted MIL CE baseline at 0.165. RankMix has the lowest raw ECE (0.151) and smallest temperature ($T \approx 1.3$) of any WSI-bag method, but its post-scaling ECE (0.144) is not the best. The elevated residual miscalibration across all BRACS methods, relative to TCGA-UT, reflects the smaller training pool and the intrinsic difficulty of seven-class fine-grained subtyping rather than a failure specific to any one method.

5.2 CAMELYON16

On CAMELYON16 the picture inverts: the same frozen features that struggle with BRACS's fine-grained subtypes separate tumor from normal tissue almost perfectly, so despite an imbalance an order of magnitude more severe, every method reaches near-ceiling discrimination in both regimes (Tables 9 and 10).

Method	Accuracy	Balanced accuracy	Macro F1
CFAL	0.997 ± 0.002	0.961 ± 0.015	0.966 ± 0.008
Balanced sampling	0.997 ± 0.001	0.969 ± 0.014	0.960 ± 0.020
ProGAN augmentation	0.996 ± 0.001	0.956 ± 0.009	0.959 ± 0.019
CE	0.996 ± 0.001	0.962 ± 0.015	0.958 ± 0.020
Focal loss	0.996 ± 0.001	0.972 ± 0.010	0.957 ± 0.027
OKO	0.996 ± 0.000	0.974 ± 0.007	0.955 ± 0.034
CE + soft F1	0.996 ± 0.001	0.986 ± 0.002	0.952 ± 0.035
Weighted CE	0.996 ± 0.002	0.966 ± 0.012	0.952 ± 0.028
CE + soft MCC	0.995 ± 0.001	0.982 ± 0.006	0.940 ± 0.050

Table 9: CAMELYON16 native patch-feature test results (tumor vs. normal tiles) for validation-selected configurations, mean ± standard deviation across patient-disjoint seeds.

In the patch regime, balanced accuracy ranges from 0.961 (CFAL) to 0.986 (CE + soft F1) and macro F1 from 0.940 to 0.966, on 99.5–99.7% accuracy. The unweighted CE baseline (BAcc 0.962, F1 0.958) sits squarely inside this band, and no mitigation method separates from it by more than the between-seed noise: CFAL edges ahead on macro F1 (0.966) and CE + soft F1 on balanced accuracy (0.986), but the spread across the whole field is under 0.03 on either

Method	Accuracy	Balanced accuracy	Macro F1
MDE-MIL	0.983 ± 0.017	0.984 ± 0.018	0.982 ± 0.018
SC-MIL	0.983 ± 0.017	0.984 ± 0.018	0.982 ± 0.018
Balanced sampling	0.977 ± 0.010	0.979 ± 0.012	0.977 ± 0.010
Focal loss	0.977 ± 0.010	0.979 ± 0.012	0.977 ± 0.010
RankMix	0.972 ± 0.020	0.974 ± 0.020	0.971 ± 0.020
MIL CE	0.966 ± 0.017	0.965 ± 0.021	0.965 ± 0.018
Weighted CE	0.966 ± 0.017	0.965 ± 0.021	0.965 ± 0.018

Table 10: CAMELYON16 native WSI-bag test results (slide-level metastasis detection) for validation-selected configurations, mean ± standard deviation across patient-disjoint seeds.

metric. The severe native imbalance (≈39:1) does not translate into a hard problem, because tumor and normal tiles are highly separable in the frozen Virchow2 space; there is little error left for any reweighting or resampling scheme to remove.

The WSI-bag regime tells the same story at the slide level. All seven methods score between balanced accuracy 0.965 and 0.984 and macro F1 0.965 and 0.982. MDE-MIL and SC-MIL lead (both F1 0.982), the unweighted MIL CE baseline (BAcc 0.965, F1 0.965, tied with Weighted CE) is again fully competitive, and the entire field is separated by less than 0.02 macro F1. Canonical slide-level metastasis detection is nearly solved from a bounded bag of Virchow2 features, leaving no meaningful gap for bag-composition or resampling methods to close.

Method	ECE	ECE+TS	T
CE	0.003 ± 0.001	0.001 ± 0.001	3.723 ± 0.142
Weighted CE	0.004 ± 0.002	0.001 ± 0.001	4.224 ± 0.503
Focal loss	0.003 ± 0.001	0.001 ± 0.000	4.547 ± 0.173
CFAL	0.464 ± 0.001	0.068 ± 0.004	0.050 ± 0.000
CE + soft F1	0.004 ± 0.001	0.001 ± 0.001	4.598 ± 0.179
CE + soft MCC	0.005 ± 0.001	0.002 ± 0.000	5.025 ± 0.191
Balanced sampling	0.003 ± 0.001	0.001 ± 0.001	4.071 ± 0.239
OKO	0.003 ± 0.000	0.001 ± 0.000	3.014 ± 0.100
ProGAN augmentation	0.003 ± 0.001	0.001 ± 0.001	4.119 ± 0.281

Table 11: CAMELYON16 native patch-feature calibration: ECE before (raw) and after (TS) temperature scaling, with the fitted temperature T, mean ± standard deviation across seeds.

Probability Calibration. Calibration follows discrimination: because the models are both confident and correct, they are well calibrated out of the box. In the patch regime raw ECE is 0.003–0.005 for every method except CFAL, and temperature scaling brings it to 0.001–0.002 (Table 11); the fitted temperatures ($T \approx$ 3–5) merely soften mildly over-confident logits. CFAL is again the lone exception, with the same near-degenerate low-temperature signature seen on BRACS (raw ECE 0.464, $T \approx 0.05$, post-scaling ECE 0.068). In the WSI-bag regime raw ECE is 0.013–0.027 and post-scaling ECE 0.017–0.034 (Table 12); MDE-MIL is the best calibrated (raw ECE 0.013), and for several methods the fitted temperature reaches the grid's lower floor ($T \approx 0.05$) without lowering ECE, reflecting how little miscalibration remains to correct on the small 398-slide cohort. Both regimes are markedly better calibrated than BRACS—patch post-scaling ECE around 0.001 versus BRACS's 0.04–0.08, and WSI-bag around 0.02–0.03 versus

Method	ECE	ECE+TS	T
MIL CE	0.027 ± 0.017	0.034 ± 0.017	0.050 ± 0.000
Weighted CE	0.027 ± 0.017	0.034 ± 0.017	0.050 ± 0.000
Focal loss	0.021 ± 0.013	0.023 ± 0.010	0.050 ± 0.000
Balanced sampling	0.022 ± 0.009	0.023 ± 0.010	0.050 ± 0.000
RankMix	0.022 ± 0.030	0.028 ± 0.024	0.273 ± 0.386
SC-MIL	0.020 ± 0.018	0.017 ± 0.017	0.050 ± 0.000
MDE-MIL	0.013 ± 0.008	0.017 ± 0.017	0.050 ± 0.000

Table 12: CAMELYON16 native WSI-bag calibration: ECE before and after temperature scaling, with the fitted temperature T. Columns as in Table 11.

BRACS's 0.11–0.15—which is the calibration counterpart of the same separability gap.

5.3 PANDA

PANDA is reported in the same two regimes on its regime-specific labels: slide-level six-class ISUP grading in the WSI-bag regime (Table 14) and tile-level binary cancer detection in the patch regime (Table 13), with calibration in Tables 15 and 16. As introduced in Section 2, ISUP grades are ordered prostate-cancer grade groups derived from Gleason morphology, with grade 0 used here for benign biopsies. The two regimes are placed deliberately on opposite sides of the difficulty axis within one cohort and under two mild native imbalances: binary cancer/benign detection is the kind of coarse tissue distinction that strong pathology features usually separate cleanly, whereas six-class ISUP grading from a bounded bag of frozen features is intrinsically hard—automated Gleason grading is a fine-grained ordinal problem on which even purpose-built, end-to-end models trained for the PANDA challenge reach only moderate agreement with expert consensus (Bulten et al., 2022). PANDA therefore serves as a within-dataset test of the report's central pattern: if class separability in the frozen representation, rather than the head-to-tail ratio, is what governs the headroom for imbalance mitigation, then one cohort with two mildly imbalanced targets should still show a large gap between an easy, separable detection regime and a hard, entangled grading regime.

In the patch regime, the mask-derived cancer/benign detection task is comparatively separable: every method reaches balanced accuracy between 0.832 and 0.860 and macro F1 between 0.811 and 0.866 (Table 13). The unweighted CE baseline sits squarely inside this band—second on macro F1 (0.859) behind only CFAL (0.866) and mid-field on balanced accuracy (0.838)—so no mitigation method separates from doing nothing by more than the between-seed noise. The methods that explicitly target the minority class buy a few points of balanced accuracy at a matching macro-F1 and accuracy cost: OKO (BAcc 0.860), focal loss (0.854), and the soft-F1 and soft-MCC objectives (both 0.852) edge above CE on balanced accuracy, while CFAL and CE lead on macro F1 and overall accuracy. ProGAN augmentation, the one synthetic-data method, lands essentially on the CE baseline (balanced accuracy 0.839, macro F1 0.855), neither helping nor hurting. The result is a tight cluster with no dominant method—the signature of a separable problem at mild imbalance.

The WSI-bag regime inverts the difficulty. Six-class ISUP grading from a bounded bag of 421 Virchow2 features is hard: macro F1 spans only 0.559–0.590 and balanced accuracy 0.565–0.588 (Table 14), roughly 0.27 below the detection regime on the very same biopsies. The unweighted MIL CE baseline (BAcc 0.566, F1 0.565) again sits within noise of the field; balanced sampling (F1 0.590), focal loss (0.583), and MDE-MIL (0.581) lead it by about 0.02, but between-seed standard deviations of 0.02–0.04 make these numerical leads rather than statistically separated outcomes, and the validation-selected weighted-CE configuration collapses exactly onto the unweighted baseline. As elsewhere in the benchmark, the method choice matters only at the margin, and no mitigation method clearly beats the no-mitigation reference.

Method	Accuracy	Balanced accuracy	Macro F1
CFAL	0.928 ± 0.004	0.832 ± 0.003	0.866 ± 0.002
CE	0.921 ± 0.004	0.838 ± 0.003	0.859 ± 0.001
ProGAN augmentation	0.918 ± 0.005	0.839 ± 0.000	0.855 ± 0.002
Weighted CE	0.915 ± 0.006	0.845 ± 0.004	0.854 ± 0.003
Focal loss	0.896 ± 0.014	0.854 ± 0.006	0.834 ± 0.012
OKO	0.892 ± 0.004	0.860 ± 0.006	0.832 ± 0.006
Balanced sampling	0.890 ± 0.004	0.848 ± 0.003	0.826 ± 0.002
CE + soft F1	0.876 ± 0.009	0.852 ± 0.006	0.813 ± 0.005
CE + soft MCC	0.874 ± 0.009	0.852 ± 0.006	0.811 ± 0.004

Table 13: PANDA native patch-feature test results (cancer vs. benign tiles) for validation-selected configurations, mean ± standard deviation across patient-disjoint seeds.

Method	Accuracy	Balanced accuracy	Macro F1
Balanced sampling	0.653 ± 0.028	0.588 ± 0.032	0.590 ± 0.033
Focal loss	0.654 ± 0.003	0.584 ± 0.016	0.583 ± 0.018
MDE-MIL	0.652 ± 0.029	0.581 ± 0.036	0.581 ± 0.040
RankMix	0.626 ± 0.014	0.571 ± 0.031	0.566 ± 0.034
MIL CE	0.639 ± 0.017	0.566 ± 0.016	0.565 ± 0.021
Weighted CE	0.639 ± 0.017	0.566 ± 0.016	0.565 ± 0.021
SC-MIL	0.619 ± 0.005	0.565 ± 0.025	0.559 ± 0.021

Table 14: PANDA native WSI-bag test results (slide-level six-class ISUP grading) for validation-selected configurations, mean ± standard deviation across patient-disjoint seeds.

Probability Calibration. Calibration follows separability, as on the other datasets. In the patch regime raw ECE is low for most methods (0.024–0.103) and temperature scaling brings every method to 0.005–0.036 (Table 15); OKO is the best calibrated (post-scaling ECE 0.005), and CFAL again shows the near-degenerate low-temperature signature seen on BRACS and CAMELYON16 (raw ECE 0.382, $T \approx 0.06$) that scaling nonetheless corrects to 0.011. In the harder WSI-bag regime raw miscalibration is substantially larger (ECE 0.129–0.252) and residual post-scaling ECE higher (0.039–0.097; Table 16), with balanced sampling the best calibrated (0.039) and MDE-MIL the worst (0.097)—the calibration counterpart of the same detection-versus-grading separability gap.

6 Discussion

Table 17 collapses the full result tables to the comparison needed for the report's central claim: the no-mitigation baseline, the best mitigation method in the same regime, and the signed macro-F1 delta between them. Figure 4 plots the same deltas against baseline macro F1, using the unweighted baseline as a coarse proxy for how separable the task already is in the frozen feature space.

Method	ECE	ECE+TS	T
CE	0.046 ± 0.007	0.035 ± 0.003	2.733 ± 0.236
Weighted CE	0.045 ± 0.005	0.036 ± 0.002	2.703 ± 0.225
Focal loss	0.024 ± 0.000	0.031 ± 0.002	1.404 ± 0.126
CFAL	0.382 ± 0.004	0.011 ± 0.003	0.060 ± 0.001
CE + soft F1	0.103 ± 0.009	0.009 ± 0.001	6.418 ± 0.324
CE + soft MCC	0.101 ± 0.008	0.008 ± 0.003	5.491 ± 0.183
Balanced sampling	0.072 ± 0.005	0.019 ± 0.003	3.642 ± 0.184
OKO	0.028 ± 0.008	0.005 ± 0.002	0.822 ± 0.027
ProGAN augmentation	0.042 ± 0.007	0.039 ± 0.002	2.595 ± 0.343

Table 15: PANDA native patch-feature calibration: ECE before (raw) and after (TS) temperature scaling, with the fitted temperature T, mean ± standard deviation across seeds.

Method	ECE	ECE+TS	T
MIL CE	0.237 ± 0.024	0.058 ± 0.013	3.410 ± 0.240
Weighted CE	0.237 ± 0.024	0.058 ± 0.013	3.410 ± 0.240
Focal loss	0.129 ± 0.022	0.062 ± 0.020	2.002 ± 0.166
Balanced sampling	0.241 ± 0.021	0.039 ± 0.009	3.723 ± 0.142
RankMix	0.144 ± 0.009	0.045 ± 0.020	1.530 ± 0.029
SC-MIL	0.252 ± 0.020	0.070 ± 0.015	3.452 ± 0.297
MDE-MIL	0.239 ± 0.035	0.097 ± 0.039	2.113 ± 0.080

Table 16: PANDA native WSI-bag calibration: ECE before and after temperature scaling, with the fitted temperature T. Columns as in Table 15.

The benchmark's central design change from the companion TCGA-UT study is the addition of an unweighted no-mitigation baseline (CE, MIL CE), and the two regimes answer the question it was added to ask quite differently. In the patch regime, most mitigation methods do not clearly separate from the CE baseline on BRACS's mild native imbalance (≈2.3:1)—Weighted CE, focal loss, CE + soft MCC, balanced sampling, and ProGAN augmentation all sit within noise of unweighted cross-entropy. Only OKO and CFAL produce a measurable gain, and even they lead by only a few points of balanced accuracy. This is a plausible consequence of the imbalance itself being mild: with a native ratio an order of magnitude below TCGA-UT's ≈28:1, there may simply be less headroom for reweighting or resampling to help. ProGAN is again informative: on TCGA-UT it led at the mildest constructed severity before collapsing as tail-class support fell to single-digit patch counts at the harshest severity. BRACS's native imbalance never reaches that tail-scarcity regime, so synthetic minority augmentation has no deficit to correct, and ProGAN lands squarely on the CE baseline (BAcc 0.460 versus 0.451), neither helping nor hurting. PANDA's cancer/benign patch regime reproduces this at a second mild native imbalance (≈4.7:1): ProGAN augmentation lands within between-seed noise of the CE baseline (macro F1 0.855 versus 0.859, balanced accuracy 0.839 versus 0.838), once more falling short of the tail-scarcity regime where synthetic minority augmentation paid off on TCGA-UT.

The WSI-bag regime tells a sharper version of the same story: the unweighted MIL CE baseline is the single best method on the balanced metrics, and every dedicated mitigation method—balanced sampling, RankMix, MDE-MIL, and most strikingly SC-MIL—scores at or *below* it. This is exactly the failure mode a no-mitigation baseline is meant to surface. Whether it reflects BRACS's smaller training pool (283 native training WSIs from 387 prepared slides, versus TCGA-UT's constructed pools of over 1000 slides), the milder native imbalance giving

Dataset	Regime	Best mitigation method	Baseline macro F1	Δ macro F1
BRACS	Patch	CFAL	0.451	+0.039
BRACS	WSI bag	Weighted CE	0.508	−0.007
CAMELYON16	Patch	CFAL	0.958	+0.008
CAMELYON16	WSI bag	MDE-MIL / SC-MIL	0.965	+0.017
PANDA	Patch	CFAL	0.859	+0.007
PANDA	WSI bag	Balanced sampling	0.565	+0.025

Table 17: Cross-dataset summary of mitigation headroom. The baseline is CE for patch regimes and MIL CE for WSI-bag regimes. Δ macro F1 is the best mitigation method's mean test macro F1 minus the corresponding no-mitigation baseline, using the values in Tables 5–14.

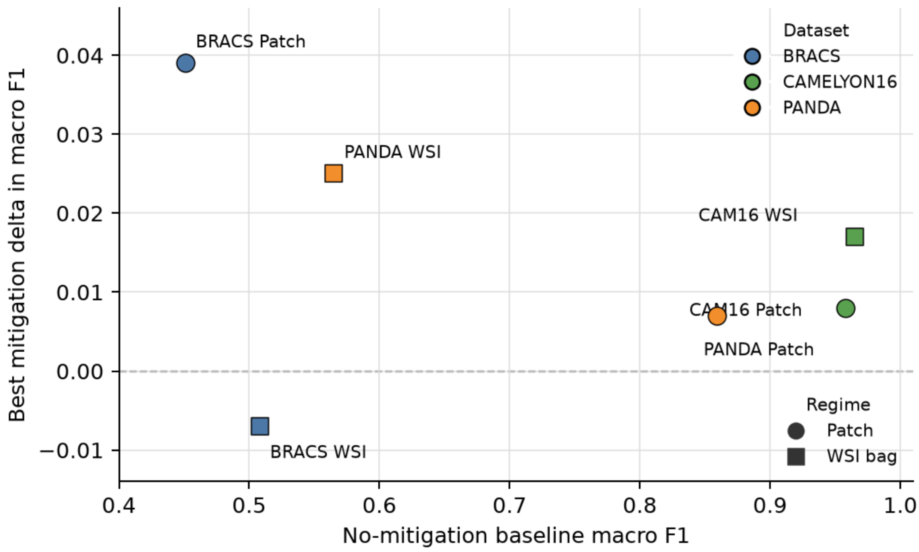


Figure 4: Relationship between no-mitigation baseline performance and mitigation headroom across all dataset–regime pairs. The vertical axis is signed: values above zero indicate that at least one mitigation method improves on the unweighted baseline, whereas values below zero indicate that every mitigation method trails it. High-baseline regimes leave little room for mitigation; lower-baseline regimes are the only settings with material positive deltas, although BRACS WSI-bag shows that low baseline performance alone does not guarantee an effective mitigation method.

resampling less to correct, or a genuine difference in how bag-composition methods behave outside a pan-cancer setting cannot be distinguished from a single additional dataset; it is a hypothesis for the next dataset in this benchmark to test.

Calibration adds a second, independent axis on which BRACS is harder than TCGA-UT: even the best-calibrated method in each regime (balanced sampling for patch, focal loss for WSI-bag) needs temperature scaling to reach an ECE that is still 2–3× worse than typical TCGA-UT post-scaling values, and the gap is especially large in the WSI-bag regime. This is consistent with the smaller training pool and finer-grained label set rather than a method-specific weakness, since the pattern holds across every method in both regimes.

CAMELYON16 sharpens the central lesson by inverting BRACS's regime. It is severely imbalanced (≈39:1, an order of magnitude beyond BRACS and comparable to TCGA-UT's ≈28:1) yet nearly solved: every method in both regimes exceeds 0.96 balanced accuracy, the unweighted baselines are competitive with the best mitigation method, and calibration is near-ideal without any dataset-specific tuning. Read together, BRACS and CAMELYON16 show that imbalance severity alone determines neither task difficulty nor the headroom available to mitigation.

A representation-level explanation is that Virchow2 appears to separate coarse, high-contrast morphology better than fine diagnostic gradations. CAMELYON16 tumor tiles differ from normal lymph-node tissue by a large tissue-state change: metastatic epithelial tumor replaces or interrupts lymphoid architecture, so the discriminative signal is broad and local enough to be captured by a frozen tile encoder. BRACS's atypical and in-situ categories are different: FEA, ADH, DCIS, and related benign proliferations lie along subtler architectural and cytologic continua inside breast epithelium, often requiring duct-level context and diagnostic thresholds that are not directly optimized by a generic frozen representation. PANDA shows the same split inside one organ: cancer versus benign tiles are comparatively separable, while six-class ISUP grading requires distinguishing ordered Gleason-pattern mixtures from bounded slide bags. The benchmark therefore exposes not only dataset difficulty, but also which kinds of pathology distinctions the frozen representation makes linearly accessible.

PANDA is included to test that separability reading with a stronger control than a cross-dataset contrast can provide. Because BRACS and CAMELYON16 differ simultaneously in imbalance ratio, organ, cohort size, and label type, a difference in mitigation headroom between them cannot be attributed to separability alone with certainty. PANDA keeps the cohort, organ, representation, and protocol fixed while comparing two mildly imbalanced targets on the same biopsies: a coarse tile-level cancer/benign detection target and a fine-grained slide-level ISUP grading target. If separability rather than the head-to-tail ratio governs the headroom for mitigation, the easy detection regime and the hard grading regime should behave differently despite coming from the same cohort. That is what the results show: the separable cancer/benign detection regime reaches balanced accuracy around 0.85, whereas the entangled six-class grading regime reaches macro F1 only around 0.58 (Tables 13 and 14), and in both regimes the unweighted baseline stays within between-seed noise of the best mitigation method. PANDA thus reproduces the separability pattern inside a single cohort, making imbalance ratio alone an implausible driver. BRACS is mildly imbalanced but hard, because fine-grained lesion subtypes are poorly separated by the frozen features; CAMELYON16 is severely imbalanced but easy, because tumor and normal tissue are nearly separable in those same features. What governs whether a mitigation method can help is therefore the separability of the classes in the representation, not the head-to-tail ratio: a severely imbalanced but separable problem leaves a plain cross-entropy baseline essentially optimal, whereas a mildly imbalanced but entangled problem is exactly where the method choice, and the no-mitigation reference, come to matter. This is a direct caution against selecting or ranking imbalance-mitigation methods by imbalance ratio alone, and it is only visible because the benchmark now spans both corners of the difficulty–imbalance plane.

7 Conclusion

This report extended the class-imbalance mitigation benchmark introduced for TCGA-UT (Queisler, 2026b) to three additional datasets on their native distributions, keeping the frozen Virchow2 representation, a bounded per-slide evidence budget, and the patient-disjoint evaluation protocol unchanged, and adding an unweighted cross-entropy baseline (CE, MIL CE) so mitigation gains are measured against doing nothing. The datasets were chosen to bracket the imbalance spectrum and to vary organ, label type, and annotation form. On BRACS's native seven-subtype distribution ($\approx$2.3:1 head-to-tail), the patch regime shows OKO and CFAL leading by a few points of balanced accuracy (BAcc 0.502 and 0.497 vs. 0.451 for CE), with every other mitigation method—including ProGAN augmentation—statistically indistinguishable from the no-mitigation baseline; in the WSI-bag regime the unweighted MIL CE baseline is the strongest method overall, and no mitigation method improves on it. On CAMELYON16's severely imbalanced patch distribution ($\approx$39:1), by contrast, every patch and WSI-bag method reaches near-ceiling discrimination (balanced accuracy 0.96–0.99) and near-ideal calibration, and the unweighted baselines are competitive with the best mitigation method in both regimes. PANDA adds a within-cohort control with the same organ, biopsies, and protocol but two different mildly imbalanced targets: the separable tile-level cancer/benign detection regime reaches balanced accuracy ≈0.85 while the entangled slide-level six-class ISUP-grading regime reaches macro F1 only ≈0.58 (Tables 13 and 14), with the no-mitigation baseline competitive in both regimes.

Taken together, the three datasets show that method rankings are not universal and, more pointedly, that imbalance severity alone does not predict whether mitigation helps: CAMELYON16 is far more imbalanced than BRACS yet leaves essentially no headroom for mitigation, because its classes are separable in the frozen representation, whereas BRACS's milder imbalance coincides with a genuinely hard, entangled task where the method choice and the no-mitigation reference matter; and PANDA reproduces the same separability effect inside a single cohort, where an easy detection regime and a hard grading regime diverge sharply while both remain only mildly imbalanced. What governs the value of imbalance mitigation under a frozen-feature benchmark is therefore class separability, not the head-to-tail ratio—a finding that only becomes visible with a native additional dataset, a genuine no-mitigation baseline, and coverage of both corners of the difficulty–imbalance plane. Further datasets can be added under the same protocol to test how far this pattern generalizes.

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